

WEEK -3

PRACTICAL

Name :- Abhay Tripathi

Roll no :- 2520030353

Sub : - DBSE&DBD

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C:\Windows\system32\cmd.e: X + v
Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\abhay>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 18
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE db;
Query OK, 1 row affected (0.03 sec)

mysql> USE db;
Database changed
mysql> CREATE TABLE roles (
  ->     role_id INT PRIMARY KEY,
  ->     role_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE users (
  ->     user_id INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     email VARCHAR(50),
  ->     role_id INT
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE departments (
```

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Query OK, 1 row affected (0.03 sec)

mysql> USE db;
Database changed
mysql> CREATE TABLE roles (
  ->     role_id INT PRIMARY KEY,
  ->     role_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE users (
  ->     user_id INT PRIMARY KEY,
  ->     name VARCHAR(50),
  ->     email VARCHAR(50),
  ->     role_id INT
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE departments (
  ->     dept_id INT PRIMARY KEY,
  ->     dept_name VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> CREATE TABLE employees (
  ->     emp_id INT PRIMARY KEY,
  ->     emp_name VARCHAR(50),
  ->     dept_id INT,
  ->     manager_id INT,
  ->     salary DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE customers (
  ->     customer_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
```

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Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE customers (
  ->     customer_id INT PRIMARY KEY,
  ->     customer_name VARCHAR(50),
  ->     city VARCHAR(50)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE orders (
  ->     order_id INT PRIMARY KEY,
  ->     customer_id INT,
  ->     order_date DATE,
  ->     total_amount DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE accounts (
  ->     account_no INT PRIMARY KEY,
  ->     holder_name VARCHAR(50),
  ->     balance DECIMAL(10,2)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql> SELECT emp_name, dept_name
  -> FROM employees e
  -> INNER JOIN departments d ON e.dept_id = d.dept_id;
Empty set (0.00 sec)

mysql> SELECT name, email, role_name
  -> FROM users u
  -> LEFT JOIN roles r ON u.role_id = r.role_id;
Empty set (0.00 sec)

mysql> SELECT name, role_name
  -> FROM users u
  -> RIGHT JOIN roles r ON u.role_id = r.role_id;
Empty set (0.00 sec)
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mysql> SELECT name, role_name
      -> FROM users u
      -> RIGHT JOIN roles r ON u.role_id = r.role_id;
Empty set (0.00 sec)

mysql> SELECT name, email, role_name FROM users u LEFT JOIN roles r ON u.role_id = r.role_id
      -> UNION
      -> SELECT name, email, role_name FROM users u RIGHT JOIN roles r ON u.role_id = r.role_id;
Empty set (0.00 sec)

mysql> ^C
mysql> SELECT e.emp_name AS Employee, m.emp_name AS Manager
      -> FROM employees e
      -> LEFT JOIN employees m ON e.manager_id = m.emp_id;
Empty set (0.00 sec)

mysql> SELECT o.order_id, customer_name, city, total_amount
      -> FROM customers c
      -> JOIN orders o ON c.customer_id = o.customer_id;
Empty set (0.01 sec)

mysql> SELECT dept_name, COUNT(*), AVG(salary), MAX(salary)
      -> FROM employees e
      -> JOIN departments d ON e.dept_id = d.dept_id
      -> GROUP BY dept_name;
Empty set (0.00 sec)

mysql> ^C
mysql> SELECT emp_name, salary, dept_name
      -> FROM employees e
      -> JOIN departments d ON e.dept_id = d.dept_id
      -> WHERE salary > 50000 AND dept_name = 'CSE';
Empty set (0.00 sec)

mysql> SELECT customer_name, SUM(total_amount) AS TotalPurchase
      -> FROM customers c
      -> JOIN orders o ON c.customer_id = o.customer_id
      -> GROUP BY customer_name
      -> ORDER BY TotalPurchase DESC;
Empty set (0.00 sec)
```

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Empty set (0.00 sec)

mysql> SELECT customer_name, SUM(total_amount) AS TotalPurchase
-> FROM customers c
-> JOIN orders o ON c.customer_id = o.customer_id
-> GROUP BY customer_name
-> ORDER BY TotalPurchase DESC;
Empty set (0.00 sec)

mysql> CREATE TABLE faculty (faculty_id INT PRIMARY KEY, faculty_name VARCHAR(50));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE courses (course_id INT PRIMARY KEY, course_name VARCHAR(50), faculty_id INT);
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE students (student_id INT PRIMARY KEY, student_name VARCHAR(50));
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE enrollments (enrollment_id INT PRIMARY KEY, student_id INT, course_id INT);
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT student_name, course_name, faculty_name
-> FROM students s
-> JOIN enrollments e ON s.student_id = e.student_id
-> JOIN courses c ON e.course_id = c.course_id
-> JOIN faculty f ON c.faculty_id = f.faculty_id;
Empty set (0.01 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES(104, 'Anil', 25000);
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES(105, 'Ravi', 10000);
```

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C:\Windows\system32\cmd.exe: X + v
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO accounts VALUES(105, 'Ravi', 10000);
Query OK, 1 row affected (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.02 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance = balance - 5000 WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> UPDATE accounts SET balance = balance + 5000 WHERE account_no = 102;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance = balance - 3000 WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> UPDATE accounts SET balance = balance + 3000 WHERE account_no = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance = balance + 2000 WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
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C:\Windows\system32\cmd.exe: X + v
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> UPDATE accounts SET balance = balance + 3000 WHERE account_no = 999;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance = balance + 2000 WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> SAVEPOINT s1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE accounts SET balance = balance - 1000 WHERE account_no = 101;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> ROLLBACK TO s1;
Query OK, 0 rows affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> |
```